Trace 2: the Disaster-FD trust-level API (colocated monitor, REST over loopback)

Targets (aggregator · peers 7 and 9) | **Disaster-FD (monitor M08)** | **ReSIDS agent (client 8)**

TL = aggregator 50 + predecessor 15 + successor 15 · TL < 30 local · 30 ≤ TL < 50 gossip · TL ≥ 50 federated · settle S = 5 s

0 min · start-up — colocated: no network discovery, the agent's endpoint is configured

agent endpoint from configuration: http://127.0.0.1:8722

GET /health →

200 {status: ok} ←

GET /status →

200 {api_version 1.9.0, current_mode local, tl_bands {gl 30, fl 50}, trust {stale}} ←

probe aggregator, peer 7, peer 9 (all isolated) ✕←

POST /trust_level {TL 0} →

204 ←

15 min · all targets reachable

probe aggregator, peer 7, peer 9 (every 5 s) ←

replies →

POST /trust_level {target aggregator, trusted, TL 80} →

204 ←

federated held 5 s → commit local → federated (architecture_change, seq 1)

30 min · aggregator lost

probe aggregator (timeout) ✕←

POST /trust_level {target aggregator, untrusted, TL 30} →

204 ←

gossip held 5 s → commit federated → gossip (seq 2)

45 min · clients 4 and 5 lost — peers of client 8 unchanged: TL stays 30, still gossip

60 min · clients 2, 3, 6 and 7 lost (region 2–7 down)

probe peer 7 (timeout) ✕←

POST /trust_level {target predecessor, untrusted, TL 15} →

204 ←

local held 5 s → commit gossip → local (seq 3; keeps detecting with its retained pool)

75 min · aggregator restored (clients 2–7 still lost)

probe aggregator ←

reply →

POST /trust_level {target aggregator, trusted, TL 65} →

204 ←

federated held 5 s → commit local → federated (seq 4)

GET /status →

200 {current_mode federated, trust {TL 65, band federated, candidate null}, last_seq 4} ←

Disaster-FD detects failures and publishes the trust level; the agent decides the mode. Its decisions are readable at GET /status and GET /events

CoAP binding: the agent subscribes instead, GET /trust_level with Observe on the monitor, and receives one notification per observation

Simulation: a trace player takes the monitor's place and issues the same POST /trust_level, replaying the Disaster-FD trace in time order